

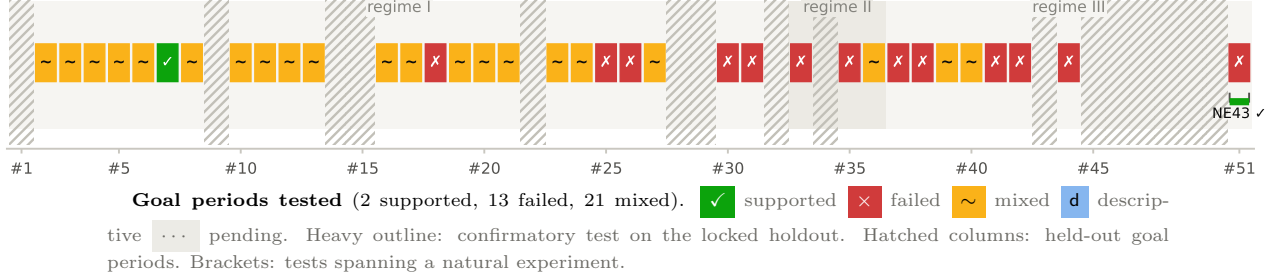
H25: A distance-to-criticality dial, T/T_c , per window and per channel

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 2/5 Usefulness 2.5/5

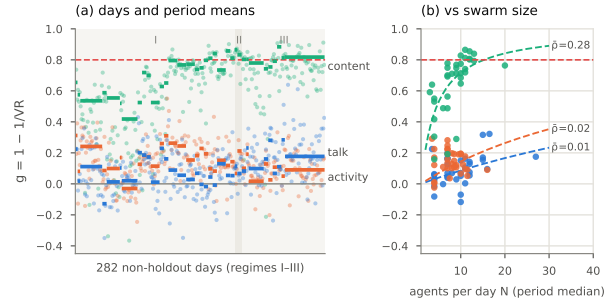
Bottom line. The daily Curie–Weiss dial puts activity and talk far below criticality ($g \approx 0.16$; activity 0.07 trimmed); near-critical content is size times a shared field.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

We build a reusable daily dial: the Curie–Weiss loop gain $g = 1 - 1/VR$ and $T/T_c = 1/g$ per day and channel, computed from logs. It masks stalls, removes slow drives and gives calibrated 90% intervals. We run it on 282 non-holdout days. **Round 1b** repeats it on the corrected activity table (round 1 lost half of all events), with the DQ8 calibrated null. **Activity and talk are subcritical on every day:** median $g = 0.16$ for both (was 0.10 and 0.12), $T/T_c \approx 6$. Trimmed to the common running window, activity drops to 0.07 and beats the independent-agent ceiling on only 24% of days. The dial now reproduces H19 on the same table ($\rho = 0.94, 0.97$) and tracks the Hawkes $\hat{n}$ of H03 ($\rho = 0.46$). **Content still reads near-critical** (0.70; P1 failed): swarm size times a shared-field correlation. Drive withdrawal (NE43) leaves the dial unchanged. Talk co-activation vanishes when 15 agents share one room (NE42). The holdout is not run.



Observables. Round 1 (old table). (a) Daily dial (dots) and period means (bars) for activity (blue), talk (orange) and content (green); red dashes mark $g = 0.8$. Activity and talk stay low; content climbs as rosters grow. On the corrected table, activity and talk shift up by about 0.06 (page 2), and content does not change. (b) Period dial vs swarm size. Dashed curves hold the per-pair correlation fixed. Content and activity follow them (field-like); talk does not (fixed table: talk $\bar{\rho} \propto N^{-1.25}$).

PHYSICAL MODEL

Curie–Weiss (01), Hawkes (09) and $O(n)$ spins (11). Spin $s_i = \pm 1$: agent i active, or posting, in a minute. For content, s_i is the 30-min mean of its whitened unit message embeddings ($n = 32$). $h_i(t)$: private fields (schedule, goal), removed by 30-min block means and agent-day means; stalls are masked. $q_i = 1 - m_i^2$. Order parameter g : loop gain. VR: collective over independent variance. $\bar{\rho}$: mean pairwise correlation. N : agents that day.

$$H = -\frac{J_0}{2N} \left(\sum_i s_i \right)^2 - \sum_i h_i(t) s_i, \quad g = \beta J_0 \bar{q}$$

$$\delta M = \frac{\sum_j q_j \beta \delta h_j}{1 - g}, \quad VR = \frac{\text{Var} \sum_i s_i}{\sum_i \text{Var} s_i} = \frac{1}{1 - g}$$

$$g = 1 - VR^{-1}, \quad T/T_c = 1/g \quad (h = 0)$$

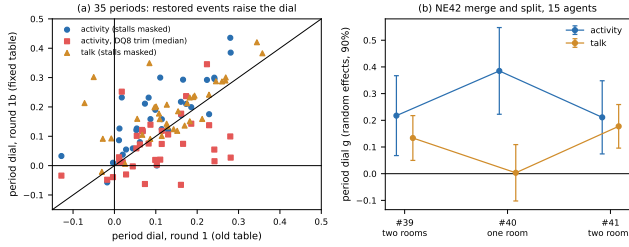
$$VR - 1 = (N - 1) \bar{\rho} \Rightarrow g(N) = \frac{(N - 1) \bar{\rho}}{1 + (N - 1) \bar{\rho}}$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Compute the dial from event logs with `compute_dial`. In this village it says the swarm amplifies a nudge by 10–20%: no fast runaway. Trim each day to the common running window before you read activity feedback. Without the trim, about half the activity “signal” is agents that boot together. Track the per-pair correlation $\bar{\rho}$ weekly, not g : at fixed $\bar{\rho}$, g rises with headcount. A green dial does not exclude slow herding through context read minutes later. The talk dial is the credible gauge: it matches an independent Hawkes fit and survives the trim.

RESULTS VS PREDICTIONS

Prediction	Round 1 → round 1b (corrected table)	Verdict
P1 every day subcritical, all channels	activity, talk 100% → 100%; content 22.5% → 24%	×
P2 medians activity 0.03–0.20, talk 0.03–0.25	0.096, 0.122 → 0.159, 0.162	✓
P3a reproduces H19 (same table)	talk ρ 0.69 → 0.97; activity 0.94 → 0.94	×
DQ8 null (trim first)	activity days above ceiling 54% new → 24%; talk 61% → 53%	
P4a talk dial vs H03 ($\rho \geq 0.3$)	$\hat{n}$ 0.25 → 0.46 (fast n_x 0.60)	×
P5 day-to-day signal ($\geq 50\%$)	2/28 → 4/29	×
P7a 03-24: activity up, talk down	+0.08/−0.06 +0.16 [0.02, 0.29]/+0.05	→ ×
NE43 drives withdrawn: no step (native)	$ z \leq 1.47$, inside placebo band	✓
NE42 room merge: size-free (native)	talk g 0.13 → 0.003 → 0.18	~



Second figure. (a) Period dials, round 1 vs 1b. Restored events raise activity and talk by about 0.06; the trim (red) pulls activity back down. (b) NE42, 15 agents: talk co-activation vanishes in the one-room week and returns at the split.

GOAL PERIODS IN DETAIL

G02 ~ 1b: mixed (round 1: mixed; corrected table, same rule)
 G03 ~ 1b: mixed (round 1: mixed; corrected table, same rule)
 G04 ~ 1b: mixed (round 1: mixed; corrected table, same rule)
 G05 ~ 1b: mixed (round 1: mixed; corrected table, same rule)
 G06 ~ 1b: mixed (round 1: mixed; corrected table, same rule)
 G07 ✓ 1b: supported (round 1: mixed; corrected table, same rule)
 ...
 G08 ~ 1b: mixed (round 1: mixed; corrected table, same rule)
 G10 ~ 1b: mixed (round 1: mixed; corrected table, same rule)
 G11 ~ 1b: mixed (round 1: mixed; corrected table, same rule)
 G12 ~ 1b: mixed (round 1: mixed; corrected table, same rule)

+26 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The 1-min dial is a lower bound, because it cannot see context-mediated coupling. The content dial is an upper bound, because shared fields survive. g depends on N . Synthetic calibration widens the intervals $\times 1.5$. The fixed-table H19 reference comes from the H38 pipeline (same estimator). The #51 size-law test is uninformative, and goals confound NE42.

ROUND 2

Add the DQ8 trim to `confirm.py` Stage B before any hold-out run. Build the lagged dial.

- **Essence.** a trustworthy number for how much the swarm amplifies a push; it must be lag-aware and size-aware, with per-pair correlation as the transferable quantity.
- **H25-R1** A lagged (DC) dial from 1–15 min cross-covariances, validated on the delayed-coupling swarms where the 1-min dial reads 0.
- **H25-R2** Agent-level availability masks from H38's scaffold states (consolidation, day edges, infra errors).
- **H25-R3** Content fields from real anchors (goal and kick-off embeddings, artifact events) plus an embedding swap; coordinate with H26.
- **H25-R4** Test the size law $g(N) = (N-1)\rho/(1+(N-1)\rho)$ at roster changes and on the holdout (CA4).
- **H25-R5** A weekly dial with change-point detection, at least 5 agents per day.